

for all cells e^I and e^J whose dimensions add up to i . Since $\Delta = \mu^*$, we have $\langle \Delta(\beta_i), e^I \otimes e^J \rangle = \langle \beta_i, \mu_*(e^I \otimes e^J) \rangle$. Because μ is the multiplication map, $\mu(e^I \times e^J)$ is contained in $P^I P^J$, and if we use the relations $P^j P^j \subset P^j P^{j-1}$ and $P^j P^k \subset P^k P^j$ for $j < k$ to rearrange the factors P^j of $P^I P^J$ so that their dimensions are in decreasing order, then the only way we will end up with a term P^i is if we start with $P^I P^J$ equal to $P^i P^0$ or $P^0 P^i$. Thus $\langle \beta_i, \mu_*(e^I \otimes e^J) \rangle = 0$ unless $e^I \otimes e^J$ equals $e^i \otimes e^0$ or $e^0 \otimes e^i$. Hence $\Delta(\beta_i)$ contains no other terms besides $\beta_i \otimes 1 + 1 \otimes \beta_i$, and β_i is primitive.

Integer Homology and Cohomology

With $\mathbb{Z}$ coefficients the homology and cohomology of $SO(n)$ turns out to be a good bit more complicated than with $\mathbb{Z}_2$ coefficients. One can see a little of this complexity already for small values of n , where the homeomorphisms $SO(3) \approx \mathbb{R}P^3$ and $SO(4) \approx S^3 \times \mathbb{R}P^3$ would allow one to compute the additive structure as a direct sum of a certain number of $\mathbb{Z}$'s and $\mathbb{Z}_2$'s. For larger values of n the additive structure is qualitatively the same:

Proposition 3D.3. $H_*(SO(n); \mathbb{Z})$ is a direct sum of $\mathbb{Z}$'s and $\mathbb{Z}_2$'s.

Proof: We compute the cellular chain complex of $SO(n)$, showing that it splits as a tensor product of simpler complexes. For a cell $e^i \subset P^{n-1} \subset SO(n)$ the cellular boundary de^i is $2e^{i-1}$ for even $i > 0$ and 0 for odd i . To compute the cellular boundary of a cell $e^{i_1} \cdots e^{i_m}$ we can pull it back to a cell $e^{i_1} \times \cdots \times e^{i_m}$ of $P^{n-1} \times \cdots \times P^1$ whose cellular boundary, by Proposition 3B.1, is $\sum_j (-1)^{\sigma_j} e^{i_1} \times \cdots \times de^{i_j} \times \cdots \times e^{i_m}$ where $\sigma_j = i_1 + \cdots + i_{j-1}$. Hence $d(e^{i_1} \cdots e^{i_m}) = \sum_j (-1)^{\sigma_j} e^{i_1} \cdots de^{i_j} \cdots e^{i_m}$, where it is understood that $e^{i_1} \cdots de^{i_j} \cdots e^{i_m}$ is zero if $i_j = i_{j+1} + 1$ since $P^{i_j-1} P^{i_j-1} \subset P^{i_j-1} P^{i_j-2}$, in a lower-dimensional skeleton.

To split the cellular chain complex $C_*(SO(n))$ as a tensor product of smaller chain complexes, let C^{2i} be the subcomplex of $C_*(SO(n))$ with basis the cells e^0 , e^{2i} , e^{2i-1} , and $e^{2i}e^{2i-1}$. This is a subcomplex since $de^{2i-1} = 0$, $de^{2i} = 2e^{2i-1}$, and, in $P^{2i} \times P^{2i-1}$, $d(e^{2i} \times e^{2i-1}) = de^{2i} \times e^{2i-1} + e^{2i} \times de^{2i-1} = 2e^{2i-1} \times e^{2i-1}$, hence $d(e^{2i}e^{2i-1}) = 0$ since $P^{2i-1} P^{2i-1} \subset P^{2i-1} P^{2i-2}$. The claim is that there are chain complex isomorphisms

$$\begin{aligned} C_*(SO(2k+1)) &\approx C^2 \otimes C^4 \otimes \cdots \otimes C^{2k} \\ C_*(SO(2k+2)) &\approx C^2 \otimes C^4 \otimes \cdots \otimes C^{2k} \otimes C^{2k+1} \end{aligned}$$

where C^{2k+1} has basis e^0 and e^{2k+1} . Certainly these isomorphisms hold for the chain groups themselves, so it is only a matter of checking that the boundary maps agree. For the case of $C_*(SO(2k+1))$ this can be seen by induction on k , as the reader can easily verify. Then the case of $C_*(SO(2k+2))$ reduces to the first case by a similar argument.

Since $H_*(C^{2i})$ consists of $\mathbb{Z}$'s in dimensions 0 and $4i-1$ and a $\mathbb{Z}_2$ in dimension $2i-1$, while $H_*(C^{2k+1})$ consists of $\mathbb{Z}$'s in dimensions 0 and $2k+1$, we conclude

from the algebraic Künneth formula that $H_*(SO(n); \mathbb{Z})$ is a direct sum of $\mathbb{Z}$'s and $\mathbb{Z}_2$'s. $\square$

Note that the calculation shows that $SO(2k)$ and $SO(2k-1) \times S^{2k-1}$ have isomorphic homology groups in all dimensions.

In view of the preceding proposition, one can get rather complete information about $H_*(SO(n); \mathbb{Z})$ by considering the natural maps to $H_*(SO(n); \mathbb{Z}_2)$ and to the quotient of $H_*(SO(n); \mathbb{Z})$ by its torsion subgroup. Let us denote this quotient by $H_*^{free}(SO(n); \mathbb{Z})$. The same strategy applies equally well to cohomology, and the universal coefficient theorem gives an isomorphism $H_{free}^*(SO(n); \mathbb{Z}) \approx H_*^{free}(SO(n); \mathbb{Z})$.

The proof of the proposition shows that the additive structure of $H_*^{free}(SO(n); \mathbb{Z})$ is fairly simple:

$$\begin{aligned} H_*^{free}(SO(2k+1); \mathbb{Z}) &\approx H_*(S^3 \times S^7 \times \cdots \times S^{4k-1}) \\ H_*^{free}(SO(2k+2); \mathbb{Z}) &\approx H_*(S^3 \times S^7 \times \cdots \times S^{4k-1} \times S^{2k+1}) \end{aligned}$$

The multiplicative structure is also as simple as it could be:

Proposition 3D.4. *The Pontryagin ring $H_*^{free}(SO(n); \mathbb{Z})$ is an exterior algebra,*

$$\begin{aligned} H_*^{free}(SO(2k+1); \mathbb{Z}) &\approx \Lambda_{\mathbb{Z}}[a_3, a_7, \dots, a_{4k-1}] \quad \text{where } |a_i| = i \\ H_*^{free}(SO(2k+2); \mathbb{Z}) &\approx \Lambda_{\mathbb{Z}}[a_3, a_7, \dots, a_{4k-1}, a_{2k+1}] \end{aligned}$$

The generators a_i are primitive, so the dual Hopf algebra $H_{free}^(SO(n); \mathbb{Z})$ is an exterior algebra on the dual generators α_i .*

Proof: As in the case of $\mathbb{Z}_2$ coefficients we can work at the level of cellular chains since the multiplication in $SO(n)$ is cellular. Consider first the case $n = 2k + 1$. Let E^i be the cycle $e^{2i}e^{2i-1}$ generating a $\mathbb{Z}$ summand of $H_*(SO(n); \mathbb{Z})$. By what we have shown above, the products $E^{i_1} \cdots E^{i_m}$ with $i_1 > \cdots > i_m$ form an additive basis for $H_*^{free}(SO(n); \mathbb{Z})$, so we need only verify that the multiplication is as in an exterior algebra on the classes E^i . The map f in the proof of Proposition 3D.2 gives a homeomorphism $e^i \times e^j \approx e^j \times e^i$ if $i < j$, and this homeomorphism has local degree $(-1)^{ij+1}$ since it is the composition $(v, w) \mapsto (v, \rho(v)w) \mapsto (v', \rho(v)w) \mapsto (\rho(v)w, v')$ of homeomorphisms with local degrees $+1, -1$, and $(-1)^{ij}$. Applying this four times to commute $E^i E^j = e^{2i}e^{2i-1}e^{2j}e^{2j-1}$ to $E^j E^i = e^{2j}e^{2j-1}e^{2i}e^{2i-1}$, three of the four applications give a sign of -1 and the fourth gives a $+1$, so we conclude that $E^i E^j = -E^j E^i$ if $i < j$. When $i = j$ we have $(E^i)^2 = 0$ since $e^{2i}e^{2i-1}e^{2i}e^{2i-1} = e^{2i}e^{2i}e^{2i-1}e^{2i-1}$, which lies in a lower-dimensional skeleton because of the relation $p^{2i}p^{2i} \subset p^{2i}p^{2i-1}$.

Thus we have shown that $H_*(SO(2k+1); \mathbb{Z})$ contains $\Lambda_{\mathbb{Z}}[E^1, \dots, E^k]$ as a subalgebra. The same reasoning shows that $H_*(SO(2k+2); \mathbb{Z})$ contains the subalgebra $\Lambda_{\mathbb{Z}}[E^1, \dots, E^k, e^{2k+1}]$. These exterior subalgebras account for all the nontorsion in $H_*(SO(n); \mathbb{Z})$, so the product structure in $H_*^{free}(SO(n); \mathbb{Z})$ is as stated.

Now we show that the generators E^i and e^{2k+1} are primitive in $H_*^{free}(SO(n); \mathbb{Z})$. Looking at the formula for the boundary maps in the cellular chain complex of $SO(n)$, we see that this chain complex is the direct sum of the subcomplexes $C(m)$ with basis the m -fold products $e^{i_1} \cdots e^{i_m}$ with $i_1 > \cdots > i_m > 0$. We allow $m = 0$ here, with $C(0)$ having basis the 0-cell of $SO(n)$. The direct sum $C(0) \oplus \cdots \oplus C(m)$ is the cellular chain complex of the subcomplex of $SO(n)$ consisting of cells that are products of m or fewer cells e^i . In particular, taking $m = 2$ we have a subcomplex $X \subset SO(n)$ whose homology, mod torsion, consists of the $\mathbb{Z}$ in dimension zero and the $\mathbb{Z}$'s generated by the cells E^i , together with the cell e^{2k+1} when $n = 2k + 2$. The inclusion $X \hookrightarrow SO(n)$ induces a commutative diagram

$$\begin{array}{ccc} H_*^{free}(X; \mathbb{Z}) & \xrightarrow{\Delta} & H_*^{free}(X; \mathbb{Z}) \otimes H_*^{free}(X; \mathbb{Z}) \\ \downarrow & & \downarrow \\ H_*^{free}(SO(n); \mathbb{Z}) & \xrightarrow{\Delta} & H_*^{free}(SO(n); \mathbb{Z}) \otimes H_*^{free}(SO(n); \mathbb{Z}) \end{array}$$

where the lower Δ is the coproduct in $H_*^{free}(SO(n); \mathbb{Z})$ and the upper Δ is its analog for X , coming from the diagonal map $X \rightarrow X \times X$ and the Künneth formula. The classes E^i in the lower left group pull back to elements we label $\tilde{E}^i$ in the upper left group. Since these have odd dimension and $H_*^{free}(X; \mathbb{Z})$ vanishes in even positive dimensions, the images $\Delta(\tilde{E}^i)$ can have no components $a \otimes b$ with both a and b positive-dimensional. The same is therefore true for $\Delta(E^i)$ by commutativity of the diagram, so the classes E^i are primitive. This argument also works for e^{2k+1} when $n = 2k + 2$.

Since the exterior algebra generators of $H_*^{free}(SO(n); \mathbb{Z})$ are primitive, this algebra splits as a Hopf algebra into a tensor product of single-generator exterior algebras $\Lambda_{\mathbb{Z}}[a_i]$. The dual Hopf algebra $H_{free}^*(SO(n); \mathbb{Z})$ therefore splits as the tensor product of the dual exterior algebras $\Lambda_{\mathbb{Z}}[\alpha_i]$, hence $H_{free}^*(SO(n); \mathbb{Z})$ is also an exterior algebra. $\square$

The exact ring structure of $H^*(SO(n); \mathbb{Z})$ can be deduced from these results via Bockstein homomorphisms, as we show in Example 3E.7, though the process is somewhat laborious and the answer not very neat.

Stiefel Manifolds

Consider the **Stiefel manifold** $V_{n,k}$, whose points are the *orthonormal k -frames* in $\mathbb{R}^n$, that is, orthonormal k -tuples of vectors. Thus $V_{n,k}$ is a subset of the product of k copies of S^{n-1} , and it is given the subspace topology. As special cases, $V_{n,n} = O(n)$ and $V_{n,1} = S^{n-1}$. Also, $V_{n,2}$ can be identified with the space of unit tangent vectors to S^{n-1} since a vector v at the point $x \in S^{n-1}$ is tangent to S^{n-1} iff it is orthogonal to x . We can also identify $V_{n,n-1}$ with $SO(n)$ since there is a unique way of extending an orthonormal $(n-1)$ -frame to a positively oriented orthonormal n -frame.

There is a natural projection $p: O(n) \rightarrow V_{n,k}$ sending $\alpha \in O(n)$ to the k -frame consisting of the last k columns of α , which are the images under α of the last k standard basis vectors in $\mathbb{R}^n$. This projection is onto, and the preimages of points are precisely the cosets $\alpha O(n-k)$, where we embed $O(n-k)$ in $O(n)$ as the orthogonal transformations of the first $n-k$ coordinates of $\mathbb{R}^n$. Thus $V_{n,k}$ can be viewed as the space $O(n)/O(n-k)$ of such cosets, with the quotient topology from $O(n)$. This is the same as the previously defined topology on $V_{n,k}$ since the projection $O(n) \rightarrow V_{n,k}$ is a surjection of compact Hausdorff spaces.

When $k < n$ the projection $p: SO(n) \rightarrow V_{n,k}$ is surjective, and $V_{n,k}$ can also be viewed as the coset space $SO(n)/SO(n-k)$. We can use this to induce a CW structure on $V_{n,k}$ from the CW structure on $SO(n)$. The cells are the sets of cosets of the form $e^J SO(n-k) = e^{i_1} \cdots e^{i_m} SO(n-k)$ for $n > i_1 > \cdots > i_m \geq n-k$, together with the coset $SO(n-k)$ itself as a 0-cell of $V_{n,k}$. These sets of cosets are unions of cells of $SO(n)$ since $SO(n-k)$ consists of the cells $e^J = e^{j_1} \cdots e^{j_\ell}$ with $n-k > j_1 > \cdots > j_\ell$. This implies that $V_{n,k}$ is the disjoint union of its cells, and the boundary of each cell is contained in cells of lower dimension, so we do have a CW structure.

Since the projection $SO(n) \rightarrow V_{n,k}$ is a cellular map, the structure of the cellular chain complex of $V_{n,k}$ can easily be deduced from that of $SO(n)$. For example, the cellular chain complex of $V_{2k+1,2}$ is just the complex C^{2k} defined earlier, while for $V_{2k,2}$ the cellular boundary maps are all trivial. Hence the nonzero homology groups of $V_{n,2}$ are

$$H_i(V_{2k+1,2}; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{for } i = 0, 4k-1 \\ \mathbb{Z}_2 & \text{for } i = 2k-1 \end{cases}$$

$$H_i(V_{2k,2}; \mathbb{Z}) = \mathbb{Z} \quad \text{for } i = 0, 2k-2, 2k-1, 4k-3$$

Thus $SO(n)$ has the same homology and cohomology groups as the product space $V_{3,2} \times V_{5,2} \times \cdots \times V_{2k+1,2}$ when $n = 2k+1$, or as $V_{3,2} \times V_{5,2} \times \cdots \times V_{2k+1,2} \times S^{2k+1}$ when $n = 2k+2$. However, our calculations show that $SO(n)$ is distinguished from these products by its cup product structure with $\mathbb{Z}_2$ coefficients, at least when $n \geq 5$, since β_1^4 is nonzero in $H^4(SO(n); \mathbb{Z}_2)$ if $n \geq 5$, while for the product spaces the nontrivial element of $H^1(-; \mathbb{Z}_2)$ must lie in the factor $V_{3,2}$, and $H^4(V_{3,2}; \mathbb{Z}_2) = 0$. When $n = 4$ we have $SO(4)$ homeomorphic to $SO(3) \times S^3 = V_{3,2} \times S^3$ as we noted at the beginning of this section. Also $SO(3) = V_{3,2}$ and $SO(2) = S^1$.

Exercises

1. Show that a topological group that has a CW structure is an orientable manifold. [Consider the homeomorphisms $x \mapsto xg$ for a fixed group element g .]
2. Using the CW structure on $SO(n)$, show that $\pi_1 SO(n) \approx \mathbb{Z}_2$ for $n \geq 3$. Find a loop representing a generator, and describe how twice this loop is nullhomotopic.
3. Compute the Pontryagin ring structure in $H_*(SO(5); \mathbb{Z})$.